

HydroSwift

Automated River-Cleaning Boat Model

Introduction: Plastic pollution in rivers is one of the leading threats to marine ecosystems, as rivers act as primary channels carrying land waste into the ocean. Addressing this problem at the river source is far more effective than cleaning open oceans. The HydroSwift project is an automated, solar-powered catamaran boat model designed to scoop up floating plastic waste from rivers before it reaches larger bodies of water.

Inspired by real-world river interceptors, HydroSwift uses a motor-driven custom conveyor belt to collect floating debris into an onboard bucket. Controlled by an Arduino Mega, the boat features an ultrasonic sensor mounted over the waste bucket to continuously measure capacity. When the bucket becomes full, the system automatically stops the conveyor motor via a relay module and flickers a warning LED to alert operators. The system also includes dedicated LED indicators to show when the conveyor is operating and when solar power is actively being harvested. Overall, HydroSwift provides an eco-friendly, efficient, and scalable prototype that demonstrates how smart automation can be deployed to keep our waterways clean.

Application Area: This embedded system utilizes Arduino-based automation, making it highly adaptable for environmental conservation groups, municipal waste management authorities, and smart-city infrastructure. It is essential to clean polluted waterways continuously, but manual river cleaning is labor-intensive, hazardous, and economically inefficient. The HydroSwift model is designed to be deployed in highly polluted local canals, rivers, and lakes.

Once deployed, the boat autonomously collects waste. Because it notifies operators visually (and can be upgraded to notify them digitally) when the bucket is full, human labor is only required for the final extraction of the waste, optimizing workforce efficiency. Furthermore, the integration of solar power extends the operational duration of the vessel, making it highly beneficial for long-term deployment in remote or off-grid areas where daily battery replacement is impractical. As a prototype, this technology serves as a proof-of-concept that can be scaled up into massive, industrial-grade interceptors capable of operating in major riverways, protecting both local marine life and global ocean ecosystems.

Technology and Tools:

- **Microcontroller:** Arduino Mega 2560
- **Distance Sensor:** Ultrasonic Sensor (HC-SR04)
- **Power Source:** 12V DC Battery & 5V/12V Solar Panel
- **Actuators:** High-Torque DC Gear Motor
- **Motor Control:** L298N Motor Driver Module (or 5V Relay Module)
- **Mechanical Parts:** Custom-built Conveyor Belt (mesh/rubber)
- **Indicators:** 3x LED Lights (Red, Yellow)
- **Structural Materials:** Marine-grade PVC pipes , PVC brackets, and acrylic sheets.
- **Miscellaneous:** Connecting wires, breadboard, soldering kit.

Language: The microcontroller is programmed using the Arduino IDE, which utilizes a derivative of C++. C++ is a powerful, human-readable, object-oriented programming language widely used in embedded systems due to its execution speed and precise hardware control. The program written in the IDE, referred to as a "sketch," is compiled into machine-readable hex code and uploaded directly to the Arduino Mega's flash memory, allowing it to execute the logic independently of a computer.

Working Mechanism of Sensors and Components:

- **Ultrasonic Sensor (HC-SR04):** The ultrasonic sensor operates on the principle of echolocation, similar to a bat or submarine sonar. It features two main components: a transmitter and a receiver. The transmitter emits a high-frequency sound wave (usually 40 kHz, inaudible to human ears). If this sound wave hits an object (in this case, the pile of plastic waste in the bucket), it bounces back as an echo and is detected by the receiver. By measuring the time it takes for the echo to return, the Arduino calculates the exact distance using the formula: $\text{Distance} = (\text{Time} \times \text{Speed of Sound in Air}) / 2$. In this project, the sensor is mounted on a tower above the bucket. When the calculated distance drops below a predefined threshold (e.g., 15 cm), the system registers that the bucket is full.

- **Solar Panel Array:** A solar panel relies on the photovoltaic effect to convert photon energy from sunlight into direct current (DC) electricity. In the HydroSwift system, the solar panel serves a dual purpose: it acts as a supplementary power source to trickle-charge the battery, and it acts as an environmental light sensor. The analog output voltage of the solar panel is connected to the Arduino's Analog-to-Digital Converter (ADC). When the voltage crosses a specific numeric threshold, the Arduino recognizes that active solar harvesting is taking place and triggers a dedicated LED to illuminate.
- **Gear Motor & Custom Conveyor Belt:** Standard DC motors spin very fast but lack the strength to pull heavy, wet garbage out of the water. Therefore, a gear motor is utilized. It uses a series of internal gears to reduce the speed (RPM) while drastically increasing the rotational force (torque). This motor drives a custom conveyor belt built from porous mesh (allowing water to drain back into the river while trapping solid plastic). The belt acts as a mechanical escalator, continuously scooping debris from the prow of the catamaran and dumping it into the central storage bucket.

Connection with ICs: The brain of the system is the Arduino Mega. The pin mapping and electrical connections are structured as follows:

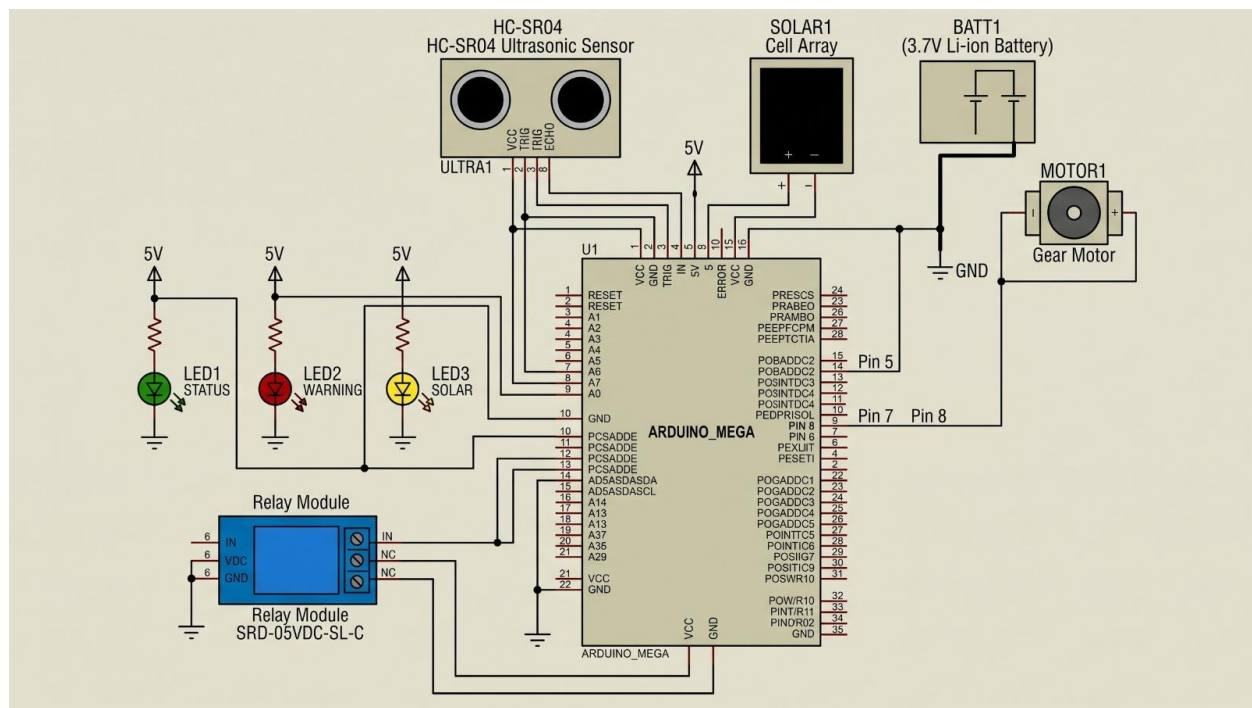
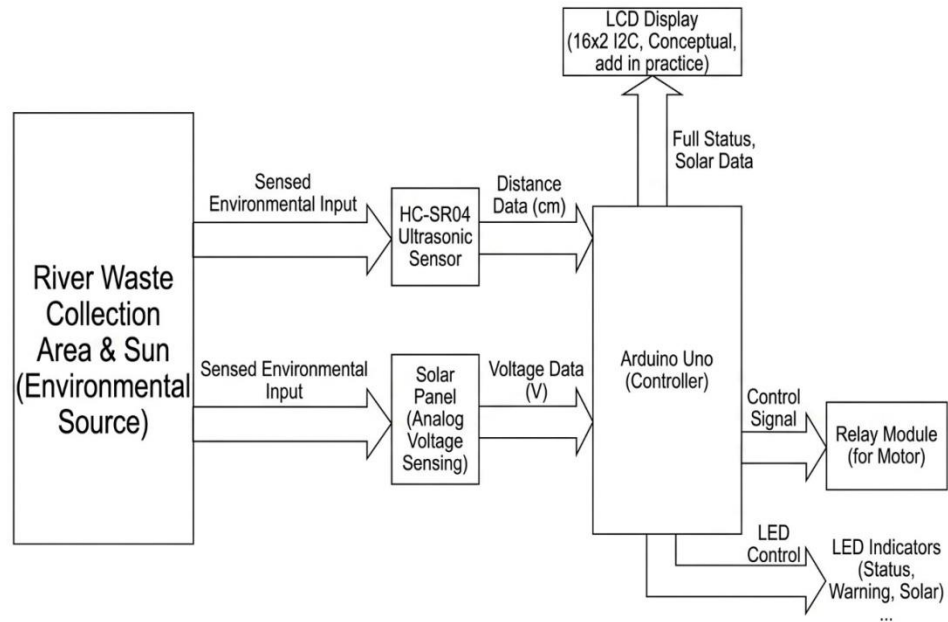


Figure: Circuit Diagram of HydroSwift

- **Pin 9 (Digital OUT):** Connected to the **TRIG** pin of the Ultrasonic sensor to emit high-frequency sound pulses.
- **Pin 10 (Digital IN):** Connected to the **ECHO** pin of the Ultrasonic sensor to listen for returning echo signals.
- **Pin 8 (Digital OUT):** Connected to the **IN / Control pin of the Relay Module**. The relay acts as an automated switch to control the high-current gear motor. (Note: Active-LOW logic is used—LOW energizes the relay to turn the motor ON, while HIGH de-energizes the relay to turn the motor OFF).
- **Pin 5 (Digital OUT):** Connected to the positive leg (anode) of **LED 1 (Status LED)** through a 220-ohm resistor. Solidly turns ON while the conveyor is actively running.
- **Pin 6 (Digital OUT):** Connected to the positive leg (anode) of **LED 2 (Warning LED)** through a 220-ohm resistor. Flickers/blinks rapidly when the ultrasonic sensor detects that the waste bucket is full.
- **Pin 7 (Digital OUT):** Connected to the positive leg (anode) of **LED 3 (Solar LED)** through a 220-ohm resistor. Illuminates when active solar panel voltage is detected.
- **Pin A0 (Analog IN):** Connected to the positive terminal of the **Solar Panel** (via a voltage divider) to measure real-time solar intensity.

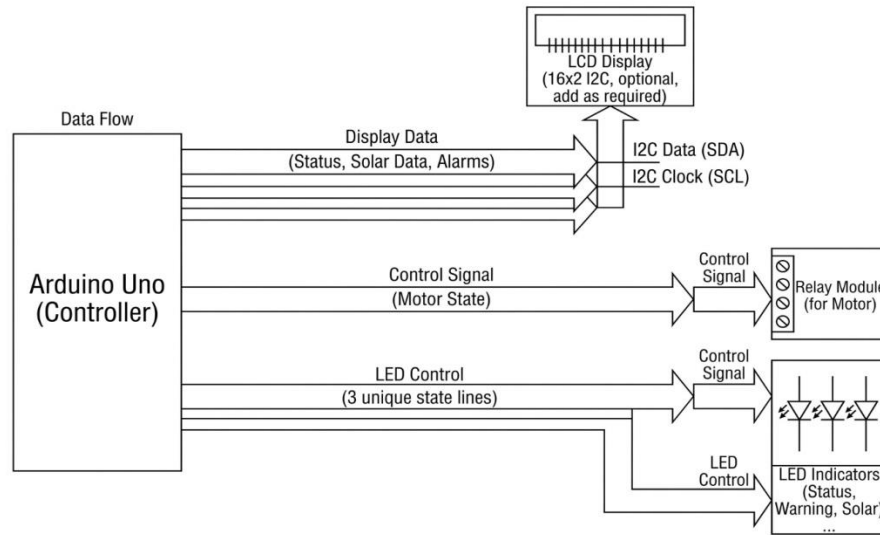
Data Flow from Sensors through ICs to I/O Devices:

- **Data Flow from Sensors to Microcontroller:** The environment provides physical inputs (sunlight and physical trash accumulation). The sensors translate these physical changes into electrical signals. The Solar panel sends a continuous varying analog voltage (0-5V) to pin A0. Simultaneously, the Arduino triggers the Ultrasonic sensor every 5 seconds. The sensor emits a pulse and sends a digital HIGH signal to pin 10 that lasts exactly as long as the sound wave took to return. The Arduino Mega's microprocessor receives these signals, digitizes them, and stores them in temporary variables for logical processing.



- **Data Flow from Microcontroller to Output Devices:** Inside the Arduino's processor, the incoming data is constantly compared against hardcoded thresholds.
 1. **Standard Operation:** If the ultrasonic distance is greater than the full threshold (meaning the bucket has space), the Arduino outputs a continuous HIGH signal to Pin 8 (keeping the motor running) and Pin 5 (keeping the operational LED on).
 2. **Interrupt Operation:** When the trash level rises and the distance drops below the threshold, the processing logic shifts. The Arduino instantly drops Pin 8 and Pin 5 to LOW, halting the motor and turning off the running LED. It then initiates an internal timer, sending alternating HIGH and LOW signals to Pin 6 every 500 milliseconds, causing the "Bucket Full" LED to blink as a distress beacon.
 3. **Parallel Processing:** Independent of the motor logic, if the Arduino reads a high integer value from the A0 analog pin, it pushes a HIGH signal to Pin 7, turning on the solar indicator LED.

HydroSweep Output Data and Control Flow Diagram



Arduino Code:

// Pin Connections

```
const int trigPin = 9;          // Ultrasonic sensor Trigger

const int echoPin = 10;         // Ultrasonic sensor Echo

const int motorPin = 8;         // Relay module IN pin

const int ledStatus = 5;        // LED 1: Solid ON when system is running (empty)

const int ledWarning = 6;       // LED 2: FLICKERS when bucket is full

const int ledSolar = 7;         // LED 3: ON when solar panel is active

const int solarPin = A0;        // Solar panel voltage sensing (Analog)
```

// Distance Configuration

```
const int fullDistanceCm = 10;  // 16cm to bottom, 6cm bucket height = 10cm threshold

const int solarThreshold = 300; // Threshold for solar activity (0-1023)
```

```
// Timing Variables
```

```
unsigned long previousSensorTime = 0;
```

```
const long sensorInterval = 3000; // 3-second interval for checking the bucket
```

```
unsigned long previousBlinkTime = 0;
```

```
const long blinkInterval = 250; // 250ms interval for a fast flicker
```

```
bool blinkState = LOW;
```

```
bool bucketIsFull = false;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(trigPin, OUTPUT);
```

```
    pinMode(echoPin, INPUT);
```

```
    pinMode(motorPin, OUTPUT);
```

```
    pinMode(ledStatus, OUTPUT);
```

```
    pinMode(ledWarning, OUTPUT);
```

```
    pinMode(ledSolar, OUTPUT);
```

```
// Initial state: Assume empty, turn ON motor (Active LOW means LOW = ON)
```

```
digitalWrite(motorPin, LOW);
```

```
digitalWrite(ledStatus, HIGH);

Serial.println("System Initialized. Waiting for 3 seconds...");

}

void loop() {

    unsigned long currentMillis = millis();

    // 1. Solar Panel Logic

    int solarValue = analogRead(solarPin);

    if (solarValue > solarThreshold) {

        digitalWrite(ledSolar, HIGH);

    } else {

        digitalWrite(ledSolar, LOW);

    }

    // 2. Ultrasonic Sensor Logic & Terminal Output

    if (currentMillis - previousSensorTime >= sensorInterval) {

        previousSensorTime = currentMillis;

        digitalWrite(trigPin, LOW);

        delayMicroseconds(2);

        digitalWrite(trigPin, HIGH);

        delayMicroseconds(10);

        digitalWrite(trigPin, LOW);

        // 30-millisecond timeout so a disconnected sensor won't freeze the board
```

```
long duration = pulseIn(echoPin, HIGH, 30000);

int distance = duration * 0.034 / 2;

// Check if distance is valid and less than or equal to 10 cm

if (distance > 0 && distance <= fullDistanceCm) {

    bucketIsFull = true;

} else {

    bucketIsFull = false;

}

// --- PRINT FULL SYSTEM STATUS TO TERMINAL ---

Serial.println("==== SYSTEM STATUS ===");

Serial.print("Sensor Distance: ");

Serial.print(distance);

Serial.println(" cm");

if (bucketIsFull) {

    Serial.println("Bucket State:  FULL");

    Serial.println("Motor State:  STOPPED");

} else {

    Serial.println("Bucket State:  HAS SPACE");

    Serial.println("Motor State:  RUNNING");

}

Serial.println("=====\n");
```

```
}
```

```
// 3. Automated Relay and LED Action Logic
```

```
if (bucketIsFull) {
```

```
    // Bucket is FULL
```

```
    digitalWrite(motorPin, HIGH);    // STOP the motor (Active LOW means HIGH = OFF)
```

```
    digitalWrite(ledStatus, LOW);    // Turn off normal status light
```

```
    // Flicker the warning light
```

```
    if (currentMillis - previousBlinkTime >= blinkInterval) {
```

```
        previousBlinkTime = currentMillis;
```

```
        blinkState = !blinkState;
```

```
        digitalWrite(ledWarning, blinkState);
```

```
    }
```

```
} else {
```

```
    // Bucket is EMPTY / Has Space
```

```
    digitalWrite(motorPin, LOW);    // RUN the motor (Active LOW means LOW = ON)
```

```
    digitalWrite(ledStatus, HIGH);  // Turn status light solidly ON
```

```
    digitalWrite(ledWarning, LOW);  // Turn off warning light
```

```
    blinkState = LOW;
```

```
}
```

```
}
```

- **Estimated Cost Analysis:** To provide a realistic budget for physical implementation, the cost of individual electronic components and structural materials has been aggregated based on standard robotics electronics retailers in Bangladesh.
- Arduino Mega 2560: 1,200 Taka
- Ultrasonic Sensor (HC-SR04): 135 Taka
- High Torque Gear Motor (12V, 60RPM): 70 Taka
- L298N Motor Driver Module: 70 Taka
- Mini Solar Panel (12V 2W): 120 Taka
- Li-Po / Lead Acid Battery (12V): 65 Taka
- LEDs (x3) and Resistors: 3 Taka
- Structural Hull Materials (PVC/Foam/Wood): 1,00 Taka
- Conveyor Materials (Rubber): 20 Taka
- Miscellaneous (Wires, Solder, Breadboard): 715 Taka

Total Estimated Cost: 2,498Taka (Note: Prices are subject to market fluctuations. Logistics, specialized tools, and waterproofing sealants are not included in this direct component estimation).

Responsibilities of Each Member:

- **Mehnaz Ahmmed:** Responsible for drafting the Introduction, defining the Application Areas, finalizing the project's conceptual framework and managing the project report.
- **Sk Yeasin Ahsanullah Al-Galib:** Tasked with researching the Working Mechanism of the sensors, defining the Connection with ICs, and creating the detailed circuit schematics.
- **Pushpita Banik** Lead on hardware fabrication. Responsible for designing the catamaran hull, building the custom conveyor belt mechanism, and mapping the data flow diagrams.
- **Aiman Al Mahmud, Aishwariya Roy** Lead embedded programmer. Responsible for writing and debugging the Arduino C++ code (implementing the `millis()` logic), calculating the Estimated Cost Analysis, and structuring the Final Conclusion.

Workplan (Gantt Chart timeline format): (To be converted into a graphical Gantt chart by the team prior to submission)

- **Week 1 (Concept & Design):** Brainstorming sessions, finalizing the HydroSwift architecture, designing the 3D/2D model of the boat, and delegating responsibilities.
- **Week 2 (Procurement & Base Assembly):** Purchasing the Arduino, sensors, and mechanical components. Constructing the buoyant catamaran base and cutting the framework for the conveyor belt.
- **Week 3 (Electronics & Programming):** Writing the C++ embedded code. Wiring the breadboard prototype to test the Ultrasonic distance calibration and the Motor Driver power delivery.
- **Week 4 (Integration & Testing):** Mounting the electronics into the hull (ensuring waterproofing). Attaching the conveyor belt to the gear motor. Conducting dry tests for the sensor logic.
- **Week 5 (Deployment & Finalization):** Conducting wet tests in a controlled body of water. Debugging mechanical friction in the conveyor and adjusting the sensor threshold. Finalizing the project report and presentation slides.

Conclusion: As urban populations grow, the stress placed on natural water systems increases exponentially. Rivers that once supported vast ecosystems are now choked with synthetic waste. The HydroSwift project demonstrates a proactive engineering response to this crisis. By transitioning away from manual, labor-heavy cleaning methods and embracing automated, sensor-driven robotics, we can address environmental cleanup with unprecedented efficiency. Our scaled model successfully proves that an autonomous vessel can harness renewable solar energy, dynamically monitor its own waste capacity using acoustic sensors, and halt operations automatically to prevent mechanical overload or garbage overflow.

While the current prototype achieves its fundamental objectives, it is a foundation with vast potential for future enhancement. The current system relies on basic visual LED indicators; however, future iterations could integrate an ESP32 WiFi module or a SIM900 GSM module to send real-time SMS or web dashboard alerts to municipal workers when the bucket is full. Additionally, integrating GPS and dual-motor differential steering could allow the vessel to

navigate river currents fully autonomously, rather than remaining stationary or tethered. Ultimately, the HydroSwift project proves that smart, scalable environmental technology is not only feasible but is an imperative step forward in restoring our natural waterways.

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